

HUANXUAN LIAO

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EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA) 2023 – Present

PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: [Shizhu He](#) and [Kang Liu](#).

North China Electric Power University (NCEPU)

2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Rank: 1/13.

RESEARCH INTERESTS

I work toward foundation models for **long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

Long-Context Data & Model Co-Design: data curation and synthesis, length- and capability-aware pretraining and mid-training mixtures, efficient attention and KV caches, and architectures that improve long-context capability, scalability, and efficiency ([LCLM-Horizon](#), [HyCo₂](#), [Spark](#), [ResAdapt](#), [AllocEmbed](#)).

Knowledge & Continual Adaptation: acquiring, transferring, internalizing and updating knowledge and capabilities, balancing plasticity, retention and generalization ([TAGI](#), [SHIP](#), [DyPRAG](#), [AAG](#), [NesyCD](#), [SKIntern](#), [SpaRTA](#)).

Agent Learning & Self-Improvement: persistent memory, long-horizon decisions and learning from interaction and feedback, toward verifiable improvement of agent memory, policies and learning procedures ([CITI](#), [KATE](#), [BuPO](#), [Tool-RL-Box](#), [OPERATE](#), [Distill Yourself](#), [SLE](#)).

SELECTED PUBLICATIONS AND WORKING PAPERS

Full publication list: [personal homepage](#).

Long-Context Data and Model Co-Design

- [13] [ACM TOIS'26] [Huanxuan Liao](#), Yao Xu, Shizhu He, Jun Zhao, and Kang Liu. “HyCo₂: Hybrid Context Compression with Global Semantics and Local Evidence”.
- [12] [arXiv'26] [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, and Kang Liu. “ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning”.
- [11] [arXiv'26] Song Jin, Zhongtao Jiang, Chenglei Shen, [Huanxuan Liao](#), Haozhe Chi, Zhiwei Wang, Kun Xu, and Yong Liu. “Allocate Before You Embed: Adaptive Visual Input Allocation for Video Embeddings”.
- [10] [AAAI'26] [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, and Kang Liu. “Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning”.
- [9] [arXiv'25] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), et al. “A Comprehensive Survey on Long Context Language Modeling”. *Core contributor to the survey.*
[Awesome-LCLM](#): **Curator & Maintainer**.

Knowledge and Continual Adaptation

- [8] [ACL'26] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, and Kang Liu. “Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs”.
- [7] [AAAI'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, and Jun Zhao. “Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks”.
- [6] [COLING'25] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, and Kang Liu. “SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models”.
- [5] [COLING'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, and Jun Zhao. “Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering”.

- [4] [NeurIPS'24] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, and Jun Zhao. "From Instance Training to Instruction Learning: Task Adapters Generation from Instructions".

Agent Learning and Self-Improvement

- [3] [EMNLP'26] Yuqiao Tan, Minzheng Wang, Shizhu He, [Huanxuan Liao](#), Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, and Kang Liu. "Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies". **Oral**.
- [2] [ACL'26] Yupu Hao, Zhuoran Jin, [Huanxuan Liao](#), Kang Liu, and Jun Zhao. "Pushing the Limits of LLM Tool Calling via Experiential Knowledge Integration and Activation". *Findings*.
- [1] [ACL'25] Yupu Hao, Pengfei Cao, Zhuoran Jin, [Huanxuan Liao](#), Yubo Chen, Kang Liu, and Jun Zhao. "Evaluating Personalized Tool-Augmented LLMs from the Perspectives of Personalization and Proactivity".

SELECTED AWARDS

- National Scholarship (three awards). 2021 / 2022 / 2025
- Beijing Outstanding Graduate. Jun. 2023

RESEARCH EXPERIENCE

Tencent (Project UP), Beijing Jul. 2026 – Present

Pretraining Data and Models

Built long-context pretraining data and tested how QA formats and data mixtures affect long-range retrieval. Evaluated sparse-attention quality and inference speed, and helped curate and clean vision-code training data.

ByteDance, Beijing Jun. 2025 – Jun. 2026

Multimodal Understanding

Deployed efficient multimodal features for AIGC detection and embedding retrieval. Proposed [ResAdapt/CAPO](#) to adjust video-frame resolution for efficient reasoning; contributed to [AllocEmbed/RDPO](#) for efficient video retrieval. Built [OPERATE](#) to evaluate agents in changing environments.

Ant Group, Beijing Feb. 2025 – May 2025

Long-Context Reasoning

Proposed [HyCo₂](#) to shorten long inputs while retaining their overall meaning and relevant evidence. Contributed to the [LCLM-Horizon survey](#) and curated and maintained [Awesome-LCLM](#).

PROFESSIONAL SERVICES

Conference Reviewer: ACL ARR 2024–2026; NeurIPS 2025–2026; ICML 2026; COLM 2026; ICLR 2025–2026; AAAI 2026.

Open Source: [Awesome-LCLM](#): **Curator & Maintainer**; [NVIDIA/kvpress](#): Contributor; [Distill Yourself](#): Core Logic & Code Contributor.